ESSAY TOPIC 2:

INVESTIGATING THE MATERIALITY OF OIL SUPPLY IN THE FORMATION OF ITS PRICE

Climate Policy & Commodity Markets

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Introduction

In a global economy that is slowly shifting towards increasingly sustainable energy sources, oil remains extremely important in sectors such as transportation, the chemical and petrochemical industries, and buildings. Its price is one of the most important economic indicators and is influenced by numerous factors beyond global supply and demand, such as market sentiment, the strength of the U.S. dollar, natural disasters, and tensions in producing countries.

This essay will analyze some of the most important research on oil price formation, investigating whether it is influenced more by demand or supply, and whether the latter is immaterial as claimed by some studies.

The case for demand dominance

Crude oil is a primary input for the global economy, this structural link is clarified by the concept of "oil intensity", which is the amount of oil required to produce a unit of GDP. As noted by Rühl and Erker (2021), the relationship between oil and GDP has evolved from intermediate industrial use to final consumption in transportation and household appliances, this shift has resulted in a linear decline in oil intensity, but also a decrease in inter-fuel substitutability, making global oil demand a sticky function of economic output (Figure 1).

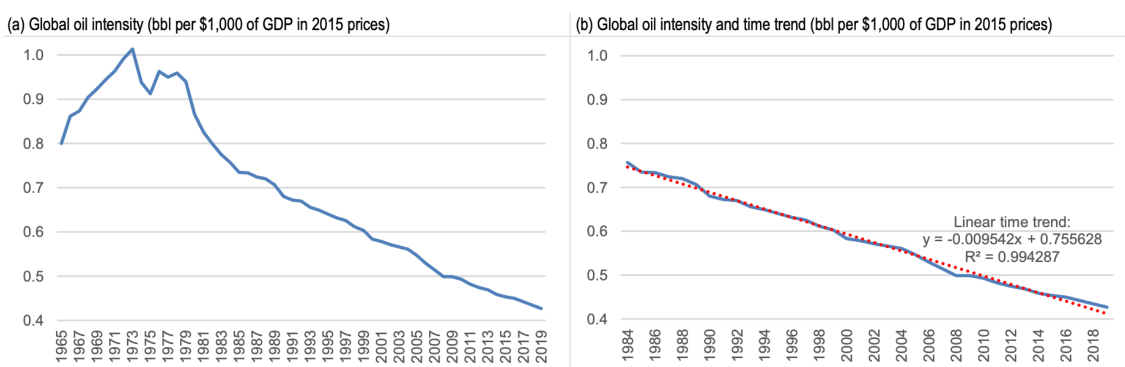

Figure 1 - Rühl and Erker (2021)

This rigidity is reflected in the IEA Oil 2025 analysis, which demonstrates that despite the ongoing energy transition, global oil demand continues to be pulled upward by the economic rise of non-OECD countries and the expansion of the petrochemical sector (Figure 2). Together, these findings suggest that the global business cycle remains the most predictable and potent force in determining the quantity of oil consumed.

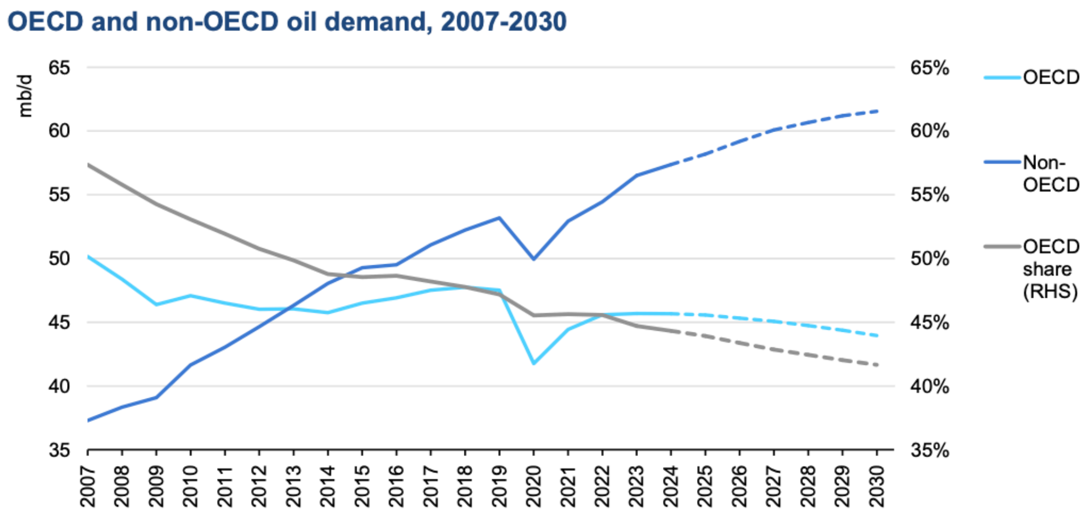

Figure 2 - Source: IEA Oil 2025

According to this logic, Kilian and Murphy (2012) developed a Structural Vector Autoregression (SVAR) model by setting several sign restrictions and physical constraints that prevent the model from providing unrealistic solutions.

The most important constraint is that on short-term supply elasticity, set at 0.0258. Other limits have been set to make the magnitude and timing of the global economic response to a change in oil prices realistic, allowing the model not to link a very small change in price to an instantaneous collapse of global demand. In addition, the model has been set up so that price shocks are consistent with the physical flow of oil (e.g., if production drops and consumption remains stable, inventories must fall).

Today's oil market is influenced by the last 24 months of history

Constant/intercept

$$B_0 y_t = \alpha + \sum_{i=1}^{24} B_i y_{t-1} + \varepsilon_t$$

Vector of orthogonal structural **innovations**: 1) flow supply shocks; 2)flow demand shock; 3) speculative demand shock

Vector of three variables: 1) %Δ in global crude oil production; 2) index of global real economic activity; 3) real price of oil

Contemporaneous **relationship** between the variables, it defines how a change in one variable affects another in the same month

Equation 1 - Kilian and Murphy (2012) VAR model

Through these constraints, Kilian and Murphy (2012) ensure that economic modeling respects geological and technical realities, effectively eliminating thousands of implausible scenarios. By doing so, they arrive at the conclusion that major historical fluctuations in oil price, such as those observed in 1979, 1991 and 2003, are primarily driven by shocks in demand. Ultimately, they argue that the view of supply shocks as major drivers of oil price volatility is fundamentally incompatible with a near zero short-run supply elasticity.

Do supply decisions matter?

The assertion that decisions regarding oil supply are “immaterial” in price formation is difficult to reconcile with the observable behavior of global energy markets. The demand-led perspective provides a solid explanation for the long-term horizon but fails to account for the volatility and “fear premiums” that characterize the market's response to changes in production.

It is essential to specify that oil supply is divided into two blocks: the "coordinated" supply managed by the OPEC+ alliance and the "price-taking" non-OPEC producers. While the core OPEC organization, a formal coalition of 12 countries including Saudi Arabia, Iran, and Venezuela, holds more than 79% of global crude reserves, its active production accounts for approximately 36% of the market. To enhance its systemic influence, the group formed the OPEC+ in 2016, a strategic alliance that includes 10 additional non-OPEC producers led by Russia. By coordinating production quotas across this expanded block, OPEC+ controls over half of the global oil supply, utilizing its collective spare capacity to actively manage prices and maintain market stability.

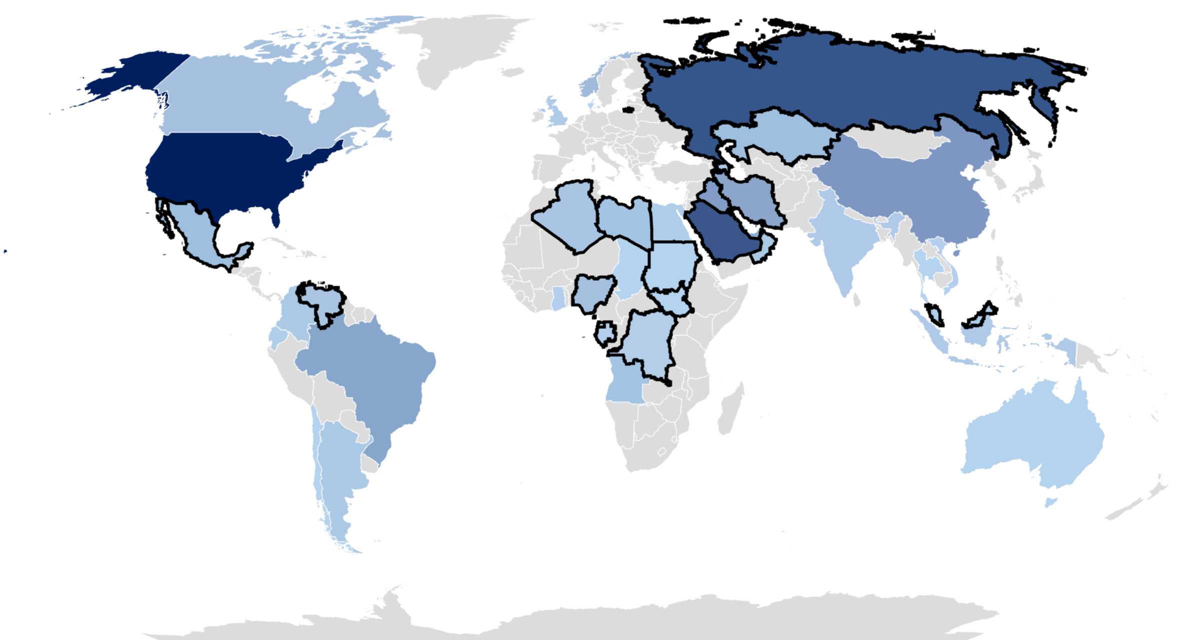

Figure 3 - World oil production by country, based on OPEC Annual Statistical Bulletin Data[1]

[1] The largest producing countries are the darkest ones, OPEC+ countries are outlined in black.

In contrast, non-OPEC supply, dominated by the United States, operates on a "price-taking basis." Producers in these regions are typically private companies rather than state-backed entities, and their output is dictated by market prices, macroeconomic conditions, and technological breakthroughs rather than centralized mandates. This is largely because oil extraction in these regions requires higher capital expenditures due to elevated exploration and production costs.

Consequently, while conventional, low-cost oil is concentrated within OPEC nations, the non-OPEC block serves as a reactive buffer to global price signals.

Together, these two forces create a supply landscape where the decision to withhold or release a marginal million barrels can be highly impactful. If supply were truly immaterial, the multi-billion dollar intelligence apparatus dedicated to monitoring OPEC spare capacity and the capital expenditure cycles of non-OPEC producers would represent a massive market inefficiency.

Often, some econometric models, such as the one considered in the previous paragraph, obscure the influence of these characteristics by relying on overly restrictive assumptions. Baumeister and Hamilton (2019) provide a seminal critique of the demand-led consensus by challenging the “physical constraints” popularized by Kilian and Murphy (2012).

The core of the new Bayesian approach is to replace the "all-or-nothing" treatment of priors with a more flexible system using probability distributions. They eventually found out that short-run elasticity of supply has a posterior median of 0.15, which is a considerably higher value than Kilian and Murphy’s estimate.

Beyond elasticity, Baumeister and Hamilton (2019) find that speculative inventory shocks played a significantly smaller role in historical price movements than often assumed. More importantly, they identify an asymmetry in macroeconomic consequences: while price increases driven by robust consumption demand are generally benign, those originating from supply disruptions exert a pronounced negative impact on subsequent economic activity. This suggests that supply-side shocks are not only material but are uniquely harmful to global growth compared to demand-driven fluctuations.

The power of news

A central flaw in the argument that supply is "immaterial" is the failure to distinguish between the physical flow of oil and the information that precedes it. In modern financial markets, the "decision" regarding supply is a material event long before the first barrel is ever pumped.

Känzig (2021) uses a new approach to analyze price changes around OPEC announcements. This is a “high-frequency identification” strategy that analyzes the “surprise” in the price of WTI futures compared to the previous day, using such a small time window that the news effect is isolated from all other variables that contribute to price formation and its expectations.

Känzig's research shows that news about future oil supply affects not only the price of oil but also its production and inventories, as well as important related macroeconomic variables such as industrial production and inflation (observed in the US).

According to the model, negative supply shock news causes an immediate rise in oil prices, which leads to a gradual but persistent decline in global oil production and an increase in inventories. This shock has stagflationary effects on the US economy, with a concomitant decline in industrial production and an increase in inflation, as well as a depreciation of the US dollar against the currencies of oil-exporting countries.

Finally, by breaking down historical oil price movements, Känzig finds that the sharp fluctuations in the late 1970s, 1985–87, 1990–91, and 2014 were largely due to news and expectations about future supply, enough to alter the global economic trajectory.

Given the close link between energy and the global economy, it is logical to interpret these decisions as vital economic signals. When OPEC announces a production cut, the market does not merely see a reduction in volume but receives an "information injection". This is the core of the “information effect” explored by Degasperi (2025), which suggests that because OPEC possesses privileged data on global consumption, its supply decisions are often interpreted as a proxy for the health of global demand.

Ultimately, if we were to accept the prompt's premise that demand is the only truly "material" force, then the market's obsession with OPEC’s supply decisions would represent an obvious logical contradiction. If supply were truly immaterial, the announcements of a supply cartel would be irrelevant noise; instead, the fact that the global economy treats these supply decisions as its primary barometer for future demand proves that supply is indispensable to process and price market information.

Is there still a fear factor?

A final key variable affecting oil prices is undoubtedly fear, meant as instability and the anticipation of adverse events such as wars, attacks, or regional tensions. The fear factor acts as a macroeconomic drag, leading to a contraction in investment and consumption. In this case supply risks are material even when no physical oil is lost, this concept was quantified by Caldara and Iacoviello (2022) through their Geopolitical Risk Index, which establishes that the mere threat of war or regional instability creates a "fear premium" that drags down global investment.

However, as oil market landscape of 2025-2026 demonstrates, the intensity of this premium is no longer a historical constant due to high global spare capacity. Industry leaders at major trading houses, including Trafigura, Vitol and Gunvor, observed that the geopolitical risk premium was dissipating because the market had become fundamentally “better able to deal with shocks” (Mitchell, S&P Global, 2025).

As Ben Luckock (Trafigura) noted, the looming surplus from producers in Guyana, Norway and Brazil created a buffer that “silenced” the traditional price reaction to conflict, this resilience proves the material force of supply in the market.

The most recent example is the US intervention in Venezuela, aimed at removing President Maduro and securing strategic access to the country with the world's largest oil reserves. The market reaction was notably muted, with only small fluctuations driven by uncertainty over the administration's next steps and the feasibility of US majors entering a market that requires deep revitalization. While the GPR index recorded a sharp increase, the same cannot be said for WTI and Brent spot prices. This divergence suggests that while the market anticipated a future production increase (and thus a price fall), the massive supply overhang currently available has effectively numbed investors to geopolitical risks.

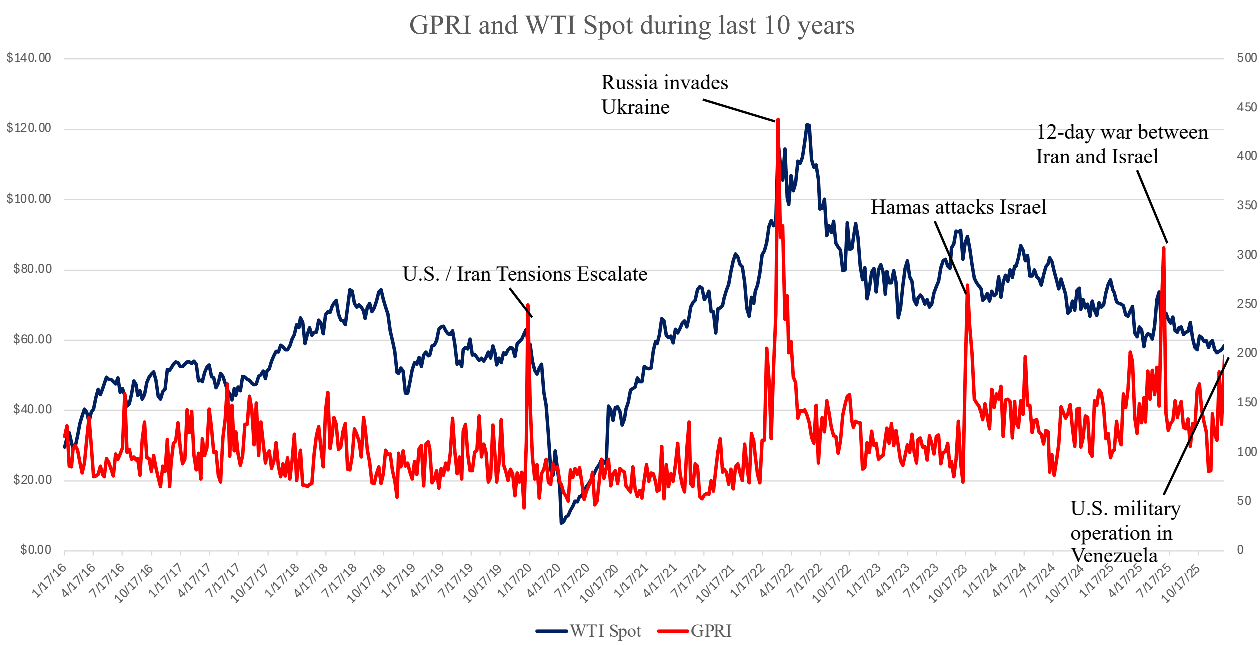

Figure 4 - Geopolitical Risk Index and WTI Spot from 2016[2]

Ultimately, the failure of this crisis to ignite a price spike does not prove that supply is immaterial, but rather that its current abundance has become the market's primary shock absorber. This “geopolitical tiredness” confirms that the material reality of a global glut now dictates the limits of fear. To ignore the supply-side is to ignore the very buffer that is currently preventing a global inflationary collapse.

[2] Data elaboration on excel from Caldara and Iacoviello (2022) and Investing.com data. (Updated 05/01/2026)

Conclusion

Analysis of the research cited leads us to refute the prompt, affirming that expectations regarding oil supply and related news are fundamental both for price formation and for the evolution of important macroeconomic indicators. Research that affirm its immateriality, such as Kilian and Murphy (2012)'s work, probably do so because they rely on assumptions that are too stringent from the outset. Finally, we can say that demand fundamentals determine the destination, but supply-side news dictates the turbulence of the journey.

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